Kinematic System of Milling Machine

Analysis of Motion Transmission and Power Flow

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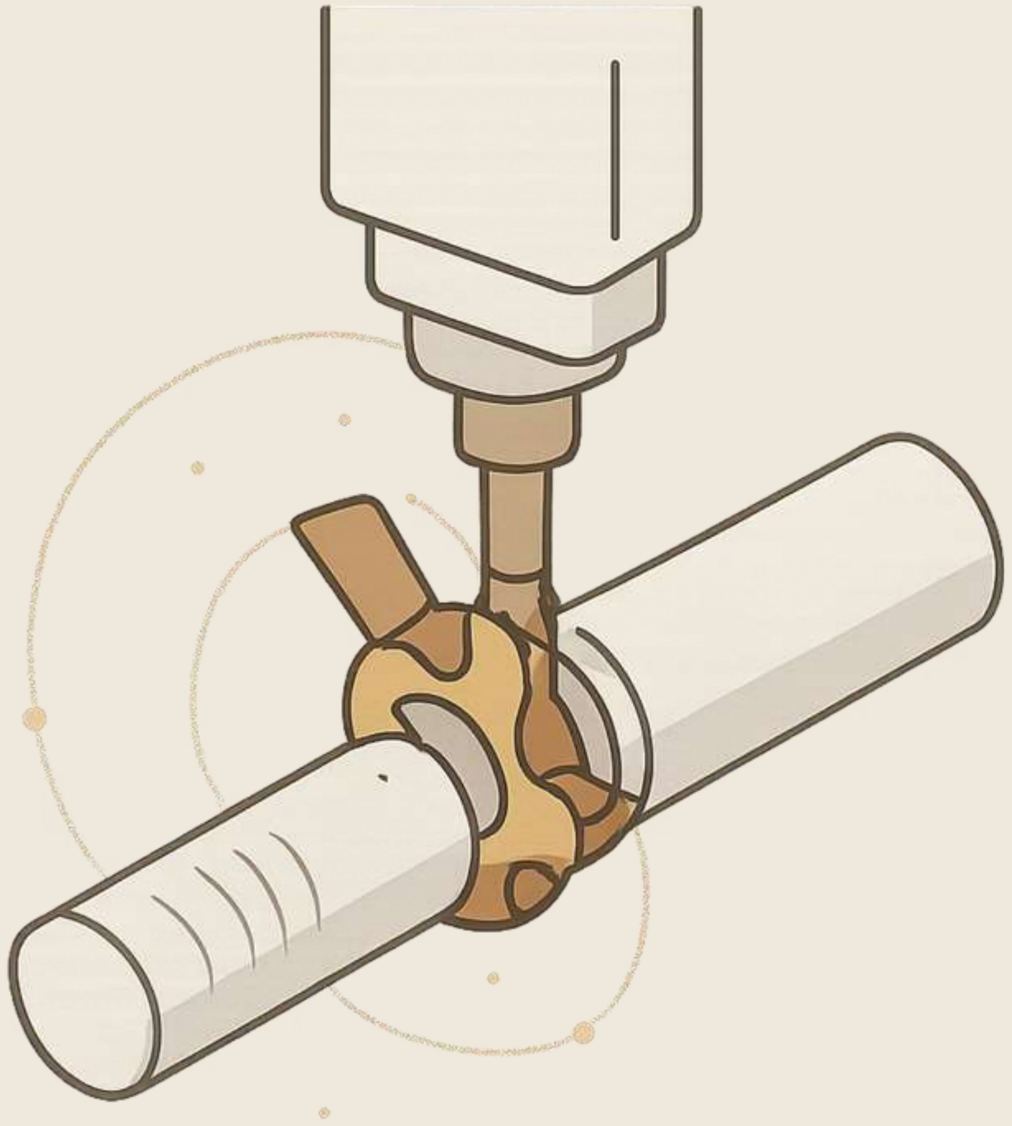
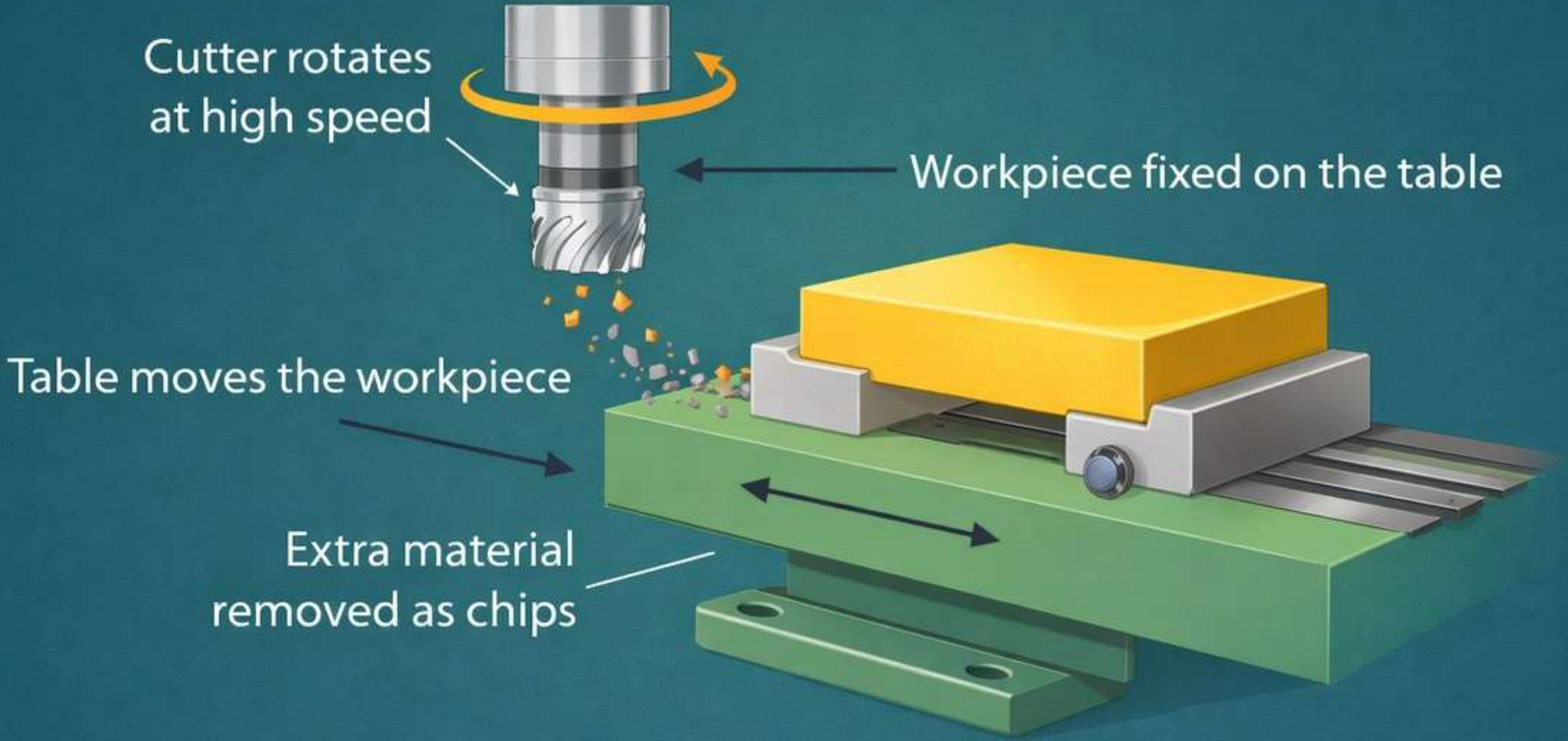
Introduction to Milling Machine and Basic Working Principle



- INTRODUCTION
- A milling machine is used to cut and shape metal or other materials.
- It removes unwanted material from the workpiece.
- A rotating cutting tool is used for cutting.
- The workpiece is moved against the cutter.
- It is widely used in workshops and industries.
- It is also used for making gears, slots, and flat surfaces.

- BASIC WORKING PRINCIPLE
- The cutter rotates at high speed.
- The workpiece is fixed on the table.
- The table moves the workpiece against the cutter.
- Extra material is removed as chips.
- The required shape and smooth surface are produced.

Basic working principle



Required shape and smooth surface produced

System Layout and Kinematic Overview

The **kinematic system** of a milling machine is its functional "nervous system" — the complete network of mechanical elements that transmit motion and power from the prime mover to all working members. A single AC motor (prime mover) drives two **parallel, independent power chains** that govern distinct machining functions.

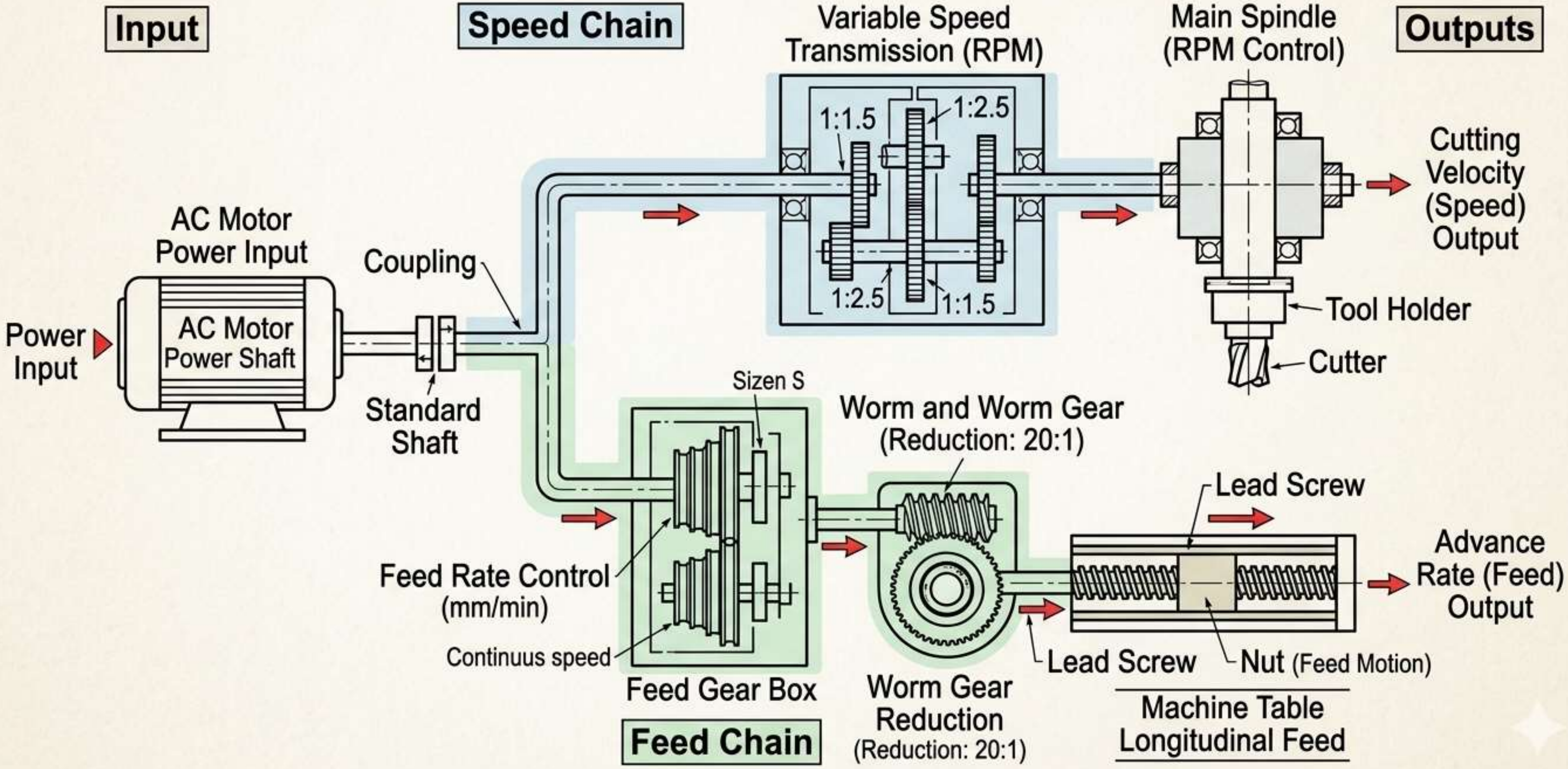
Speed Chain

Transmits rotational power to the **main spindle**, determining cutter speed (RPM). Governed by the speed gearbox and sliding gear clusters.

Feed Chain

Transmits controlled linear motion to the machine table, governing workpiece advancement rate (mm/min). Regulated by the feed gearbox and lead screw mechanism.

Kinematic Link of a Vertical Milling Machine - Drive System Analysis



Fundamental Concepts of Kinematic Chains

A **kinematic chain** is an assembly of rigid links (shafts, gears, bearings) connected by kinematic pairs, transmitting definite motion from input to output. In machine tools, these chains must maintain **mechanical rigidity** to resist intermittent cutting forces without deflection.

Prismatic Pairs

Sliding gears mounted on **splined shafts** allow axial translation to engage different gear ratios. This enables discrete speed selection without interrupting power flow — a critical feature of stepped gearboxes.

Turning Pairs

Shafts rotating within **high-precision anti-friction bearings** (ball or roller). These pairs define the rotational degrees of freedom and must minimize runout to maintain dimensional accuracy.

Structural Rigidity

The chain must resist **intermittent cutting forces** that induce shock loading. Stiffness is ensured through preloaded bearings, rigid housing, and short overhangs to prevent chatter and maintain tolerances.

Analysis of Primary Motion — Cutting Motion

Definition & Transmission Path

The **primary (cutting) motion** is the continuous rotation of the spindle and cutter about its axis. This is the motion responsible for material removal. The power transmission path is:

AC Motor → Speed Gearbox (ratio selection)

→ Main Spindle (rotational output)

→ Milling Cutter (cutting action)

Cutting Velocity Formula

The cutting velocity *V_c* (m/min) is defined as:

$$V_c = \frac{\pi \quad D \quad n}{1000}$$

Where *D* = cutter diameter (mm), *n* = spindle speed (RPM).

Effect on Tool Life: Higher *V_c* increases heat generation at the cutting zone, accelerating tool wear. Optimal *V_c* balances productivity with tool economics.

Spindle Speed Regulation — Geometric Progression

Spindle speeds in a milling machine are not arranged arbitrarily — they follow a **Geometric Progression (G.P.)** to ensure uniform percentage increments between successive speed steps, optimizing machine utility across all cutter diameters.

Common Ratio (φ)

The standard common ratio *phi* takes preferred values from the Renard series: **1.26, 1.41, or 1.58**. Each successive speed is:

$$n_k = n_1 \quad \phi^{(k-1)}$$

This ensures that the **percentage change** in cutting velocity remains constant, regardless of the speed range selected.

Engineering Rationale

- A constant *phi* ensures optimal utilization of the full speed range for **all cutter diameters**
- Minimizes the number of speed steps while maintaining adequate coverage
- Sliding gear clusters (2 or 3-stage) implement the G.P. mechanically

Analysis of Feed Motion — Workpiece Advancement

The **feed motion** is the controlled relative advancement of the workpiece against the rotating cutter. It governs chip thickness, surface finish, and machining time. Feed is quantified at three hierarchical levels:

1

Feed per Tooth (*f_z*)

The distance the workpiece advances per **individual cutting edge** engagement. Units: mm/tooth. This is the fundamental parameter for **chip load calculation** and tool life prediction.

2

Feed per Revolution (*f_{rev}*)

Total advancement per full spindle revolution. Computed as:

$$f_{rev} = f_z \times z$$

Where *z* = number of cutter teeth. Governs surface roughness profile.

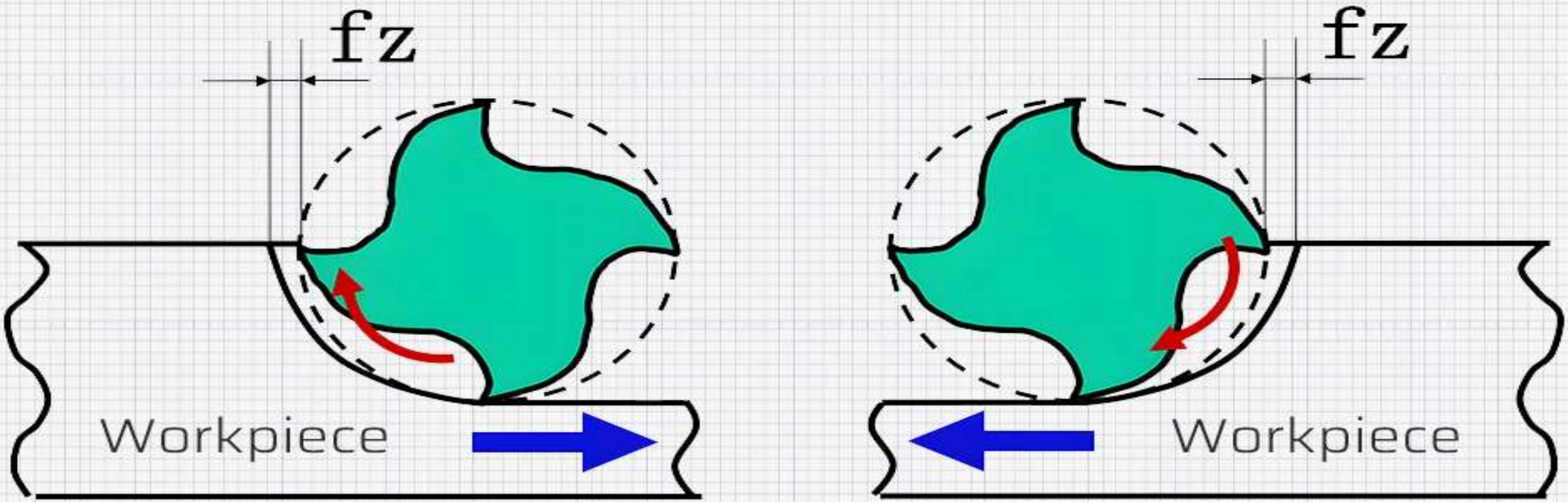
3

Table Feed Rate (*v_f*)

Linear velocity of the table in mm/min. The primary machine setting:

$$v_f = f_z \quad z \quad n \quad (\mathrm{mm/min})$$

Directly controls machining time and must be synchronized with *V_c*.



Up Milling / Climb Milling

Down Milling / Conventional Milling

KINEMATIC ANALYSIS

3-Axis Cartesian Coordinate System

The milling machine's kinematic architecture is built upon a **3-axis Cartesian coordinate system**, enabling full spatial positioning of the workpiece relative to the rotating cutter. Each axis is driven by an independent lead screw and guideway assembly, and their **simultaneous actuation** enables complex 3D contouring.

X-Axis — Longitudinal

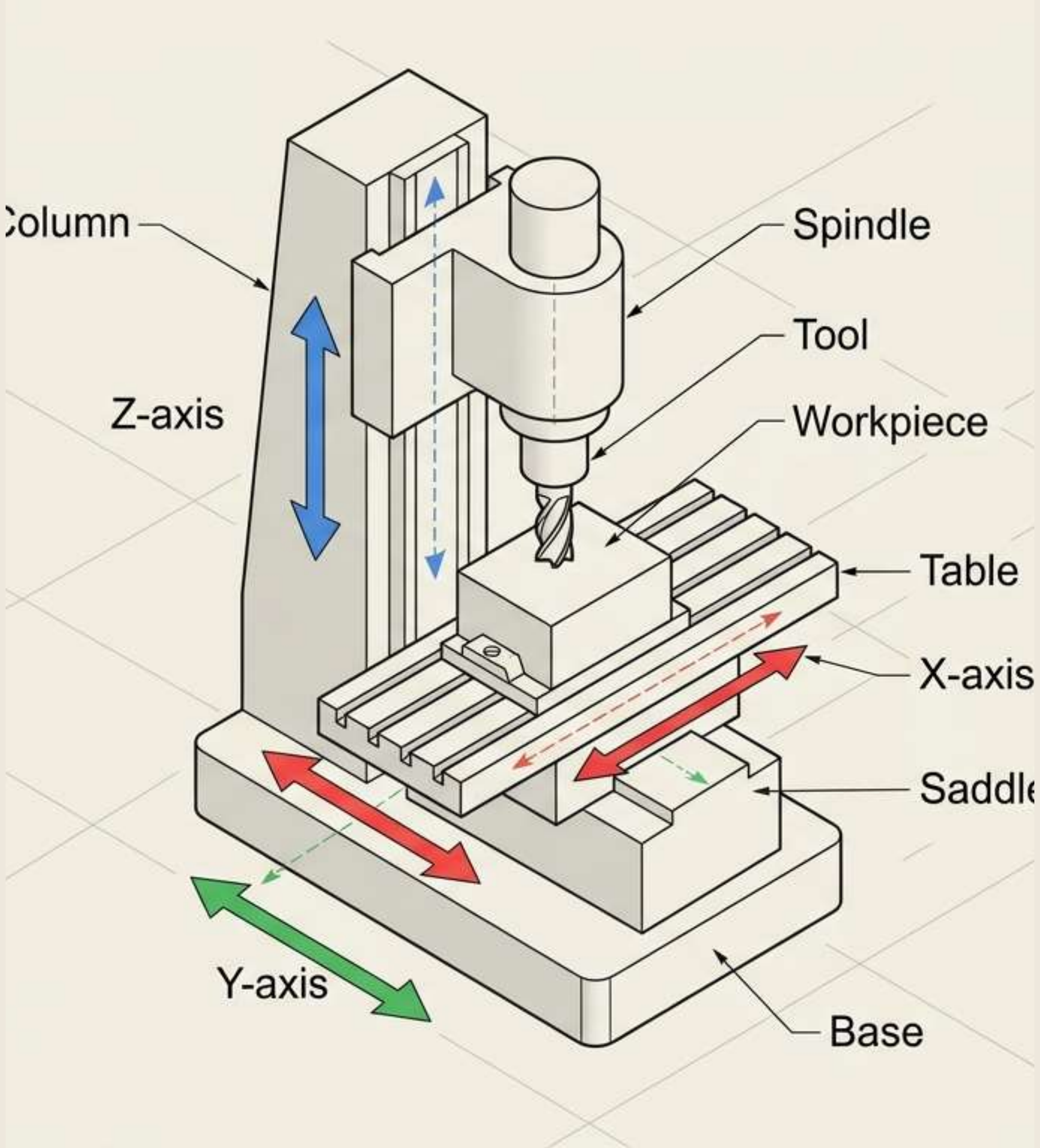
Left-to-right table travel along the **saddle guideways**. Primary axis for linear cuts and slotting. Driven by the longitudinal feed lead screw.

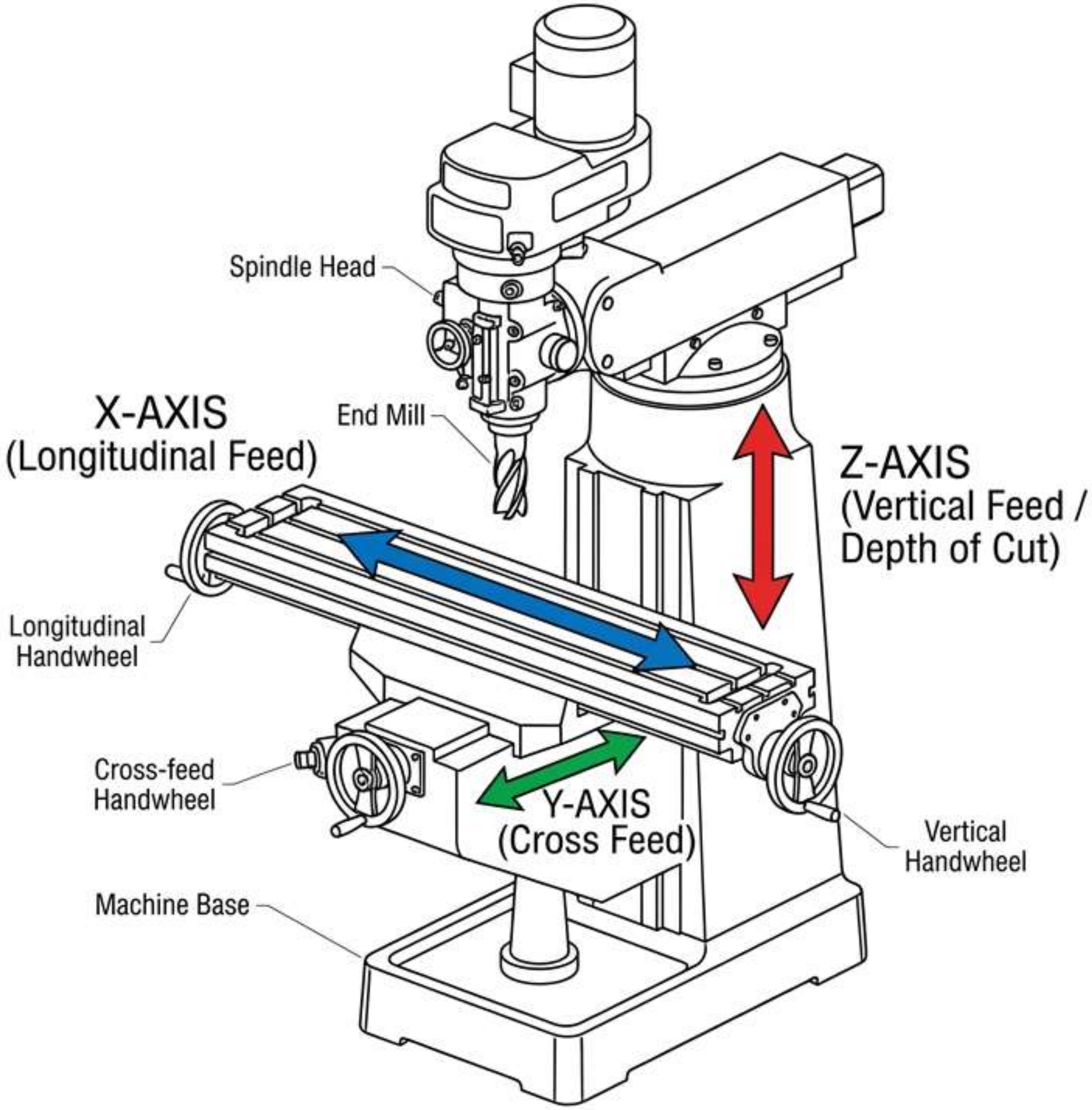
Y-Axis — Cross

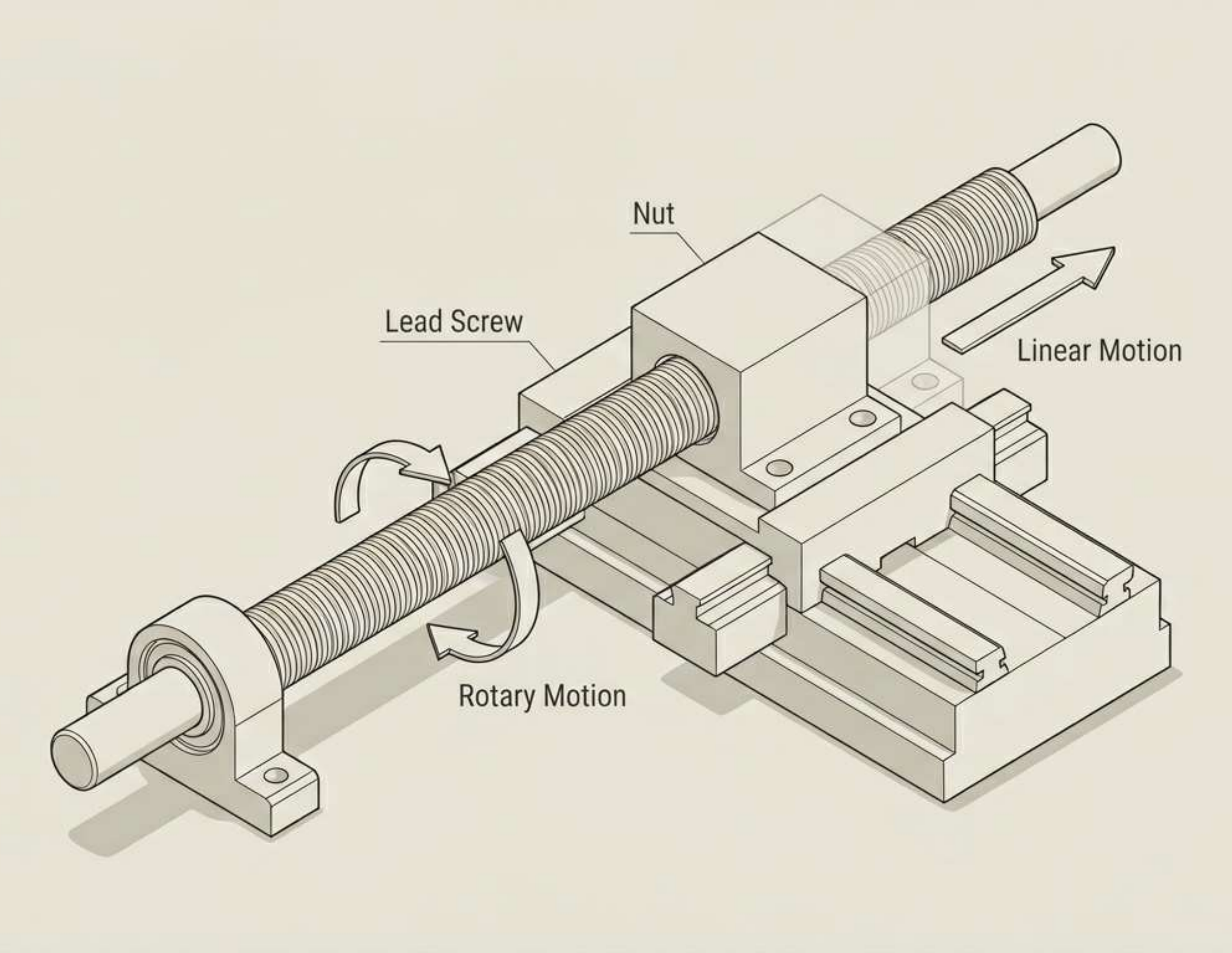
In-to-out saddle movement along the **knee guideways**. Controls depth of engagement and cross-feed. Perpendicular to X-axis.

Z-Axis — Vertical

Up-down knee movement along the **column face guideways**. Sets cutting depth and controls vertical approach. Driven by the vertical elevating screw.




Spindle Head
X-AXIS
(Longitudinal Feed)
End Mill
Z-AXIS
(Vertical Feed /
Depth of Cut)
Longitudinal
Handwheel
Cross-feed
Handwheel
Y-AXIS
(Cross Feed)
Vertical
Handwheel
Machine Base



KINEMATIC ANALYSIS

Rotary-to-Linear Conversion: The Lead Screw & Nut

The most critical kinematic transformation in a milling machine is the conversion of **rotational torque** from the feed drive into **precise linear table travel**. This is accomplished through the interaction of a hardened steel **Lead Screw** and a **Bronze Nut** fixed to the table saddle. The differential hardness — steel screw against softer bronze nut — ensures wear is concentrated on the replaceable nut, preserving the more costly screw.

KINEMATIC ANALYSIS

Pitch, Displacement & Backlash Elimination

Lead Screw Pitch & Displacement Accuracy

The **Lead Screw Pitch** defines linear displacement per revolution: one full rotation advances the table by the pitch distance. A 5 mm pitch screw produces exactly 5 mm of travel per revolution. The relationship between feed rate and spindle speed is:

$$v_f = n \times p$$

where *v_f* = feed rate (mm/min), *n* = screw RPM, and *p* = pitch (mm/rev).

Backlash Elimination

Backlash — axial play between screw and nut threads — causes lost motion during direction reversal, introducing positional errors. Elimination methods include:

- **Split-Nut Design:** Two nut halves preloaded by a spring to take up clearance
- **Preloaded Ball Screws:** Recirculating ball elements with controlled axial preload
- **Dual-Nut Preload:** Two nuts offset axially to eliminate free play

KINEMATIC ANALYSIS

Auxiliary and Indexing Mechanisms

Rapid Traverse Mechanism

The **Rapid Traverse** provides high-speed table positioning that **bypasses the feed gearbox**, engaging the lead screw directly at maximum motor RPM. This reduces non-cutting setup time and is disengaged before cutting to prevent damage.

Universal Dividing Head

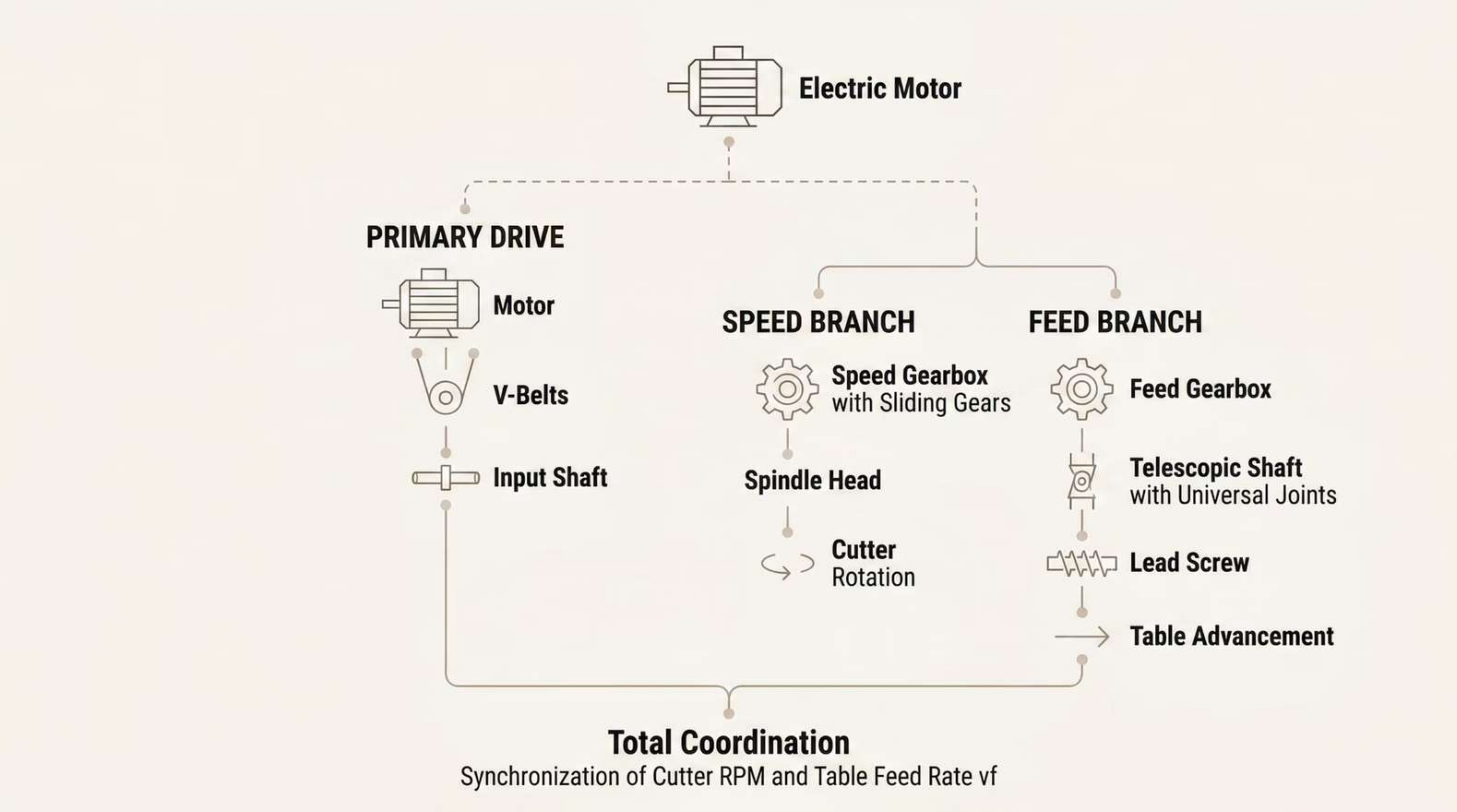
A precision kinematic attachment for angular indexing of gears, splines, and fluted cutters. Its core is a **worm and worm-gear set** with a *40:1* reduction ratio:

$$N = \frac{40}{n}$$

where N = crank turns and n = divisions. For 20 divisions, exactly 2 crank turns are required per indexed position.

KINEMATIC SYNTHESIS

Complete Power Path Synthesis



THANK YOU